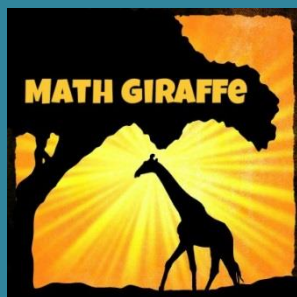


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TWO-COLUMN
PROOFS
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- Brigid

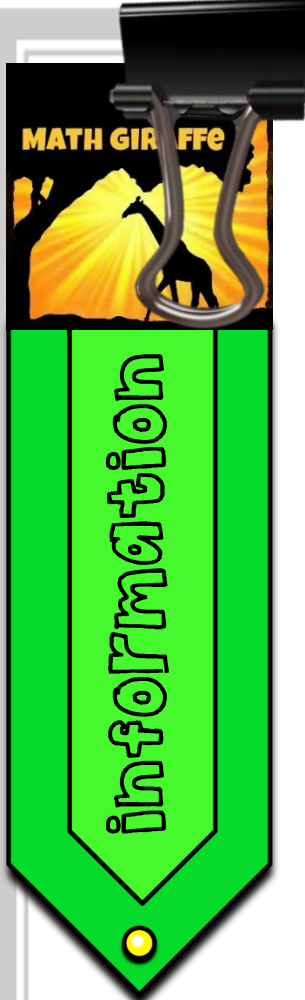


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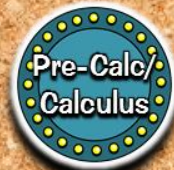
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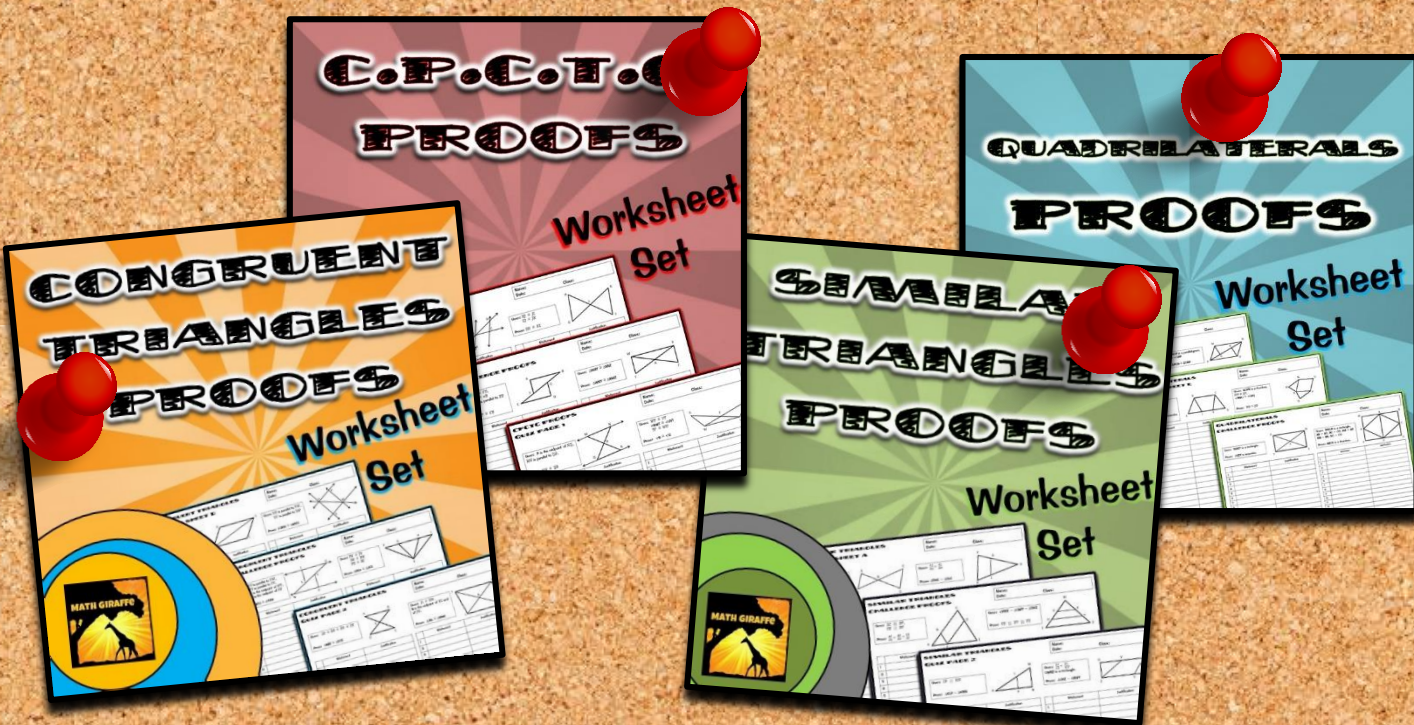
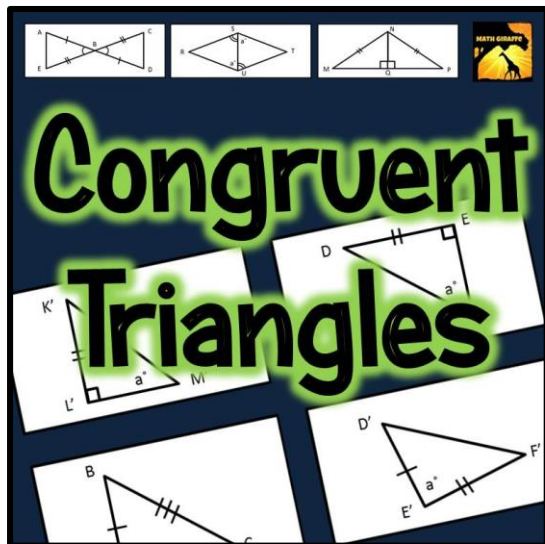
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GRAPHIC ORGANIZER: PROPERTIES OF EQUALITY AND CONGRUENCE

Name: _____

Date: _____ Class: _____

Addition Property of Equality

If $a = b$, then
 $a + x = b + x$

Subtraction Property of Equality

If $a = b$, then
 $a - x = b - x$

Multiplication Property of Equality

If $a = b$, then
 $ax = bx$

Division Property of Equality

If $a = b$, then
 $\frac{a}{x} = \frac{b}{x}$

BASIC PROPERTIES OF EQUALITY

(For all nonzero real numbers)



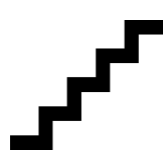
Reflexive Property

$$a = a$$



Symmetric Property

If $a = b$,
Then $b = a$



Transitive Property

If $a = b$ and $b = c$,
Then $a = c$



Substitution Property

If $a = b$, then a can
replace b in an
equation.

EQUALITY

EQUALITY

EQUALITY

EQUALITY

CONGRUENCE

CONGRUENCE

CONGRUENCE

CONGRUENCE

$$\angle A \cong \angle A$$

Or

$$\overline{AB} \cong \overline{AB}$$

If $\angle A \cong \angle B$,
then $\angle B \cong \angle A$

Or

If $\overline{AB} \cong \overline{CD}$,
Then $\overline{CD} \cong \overline{AB}$

If $\angle A \cong \angle B$ and
 $\angle B \cong \angle C$, then $\angle A \cong \angle C$

Or

If $\overline{AB} \cong \overline{CD}$ and
 $\overline{CD} \cong \overline{EF}$, then
 $\overline{AB} \cong \overline{EF}$

If $\angle A \cong \angle B$, then $\angle A$
can replace $\angle B$ in a
congruency statement.

Or

If $\overline{AB} \cong \overline{CD}$, then
 $\overline{AB}$ can replace $\overline{CD}$
in a congruency
statement.

PROPERTIES OF EQUALITY AND CONGRUENCE



PROPERTIES OF EQUALITY AND CONGRUENCE

Name: _____
Date: _____ Class: _____

Complete the diagrams. The boxes should be filled in with equations and congruency statements. The arrows should be filled in with the properties needed to get from one step to another.

$$14m + 12 = 44 - 2m$$

Addition Property of Equality

$$14m + 2m + 12 = 44 - 2m + 2m$$

$$\text{or } 16m + 12 = 44$$

1

$$\frac{16m}{16} = \frac{32}{16} \text{ or } m = 2$$

$$\angle TUV \cong \angle VWX,$$

$$\angle TUV \cong \angle PQR$$

Transitive Property of Congruence

Symmetric Property of Congruence

2

$$m\angle 2 = m\angle 3,$$

$$m\angle 3 = m\angle 4$$

$$m\angle 3 + m\angle 5 = m\angle 4 + m\angle 5$$

3

$$m\angle 2 + m\angle 5 = m\angle 4 + m\angle 5$$

Subtraction Property of Equality

In diagram 3, which property could get from the first step to the last more quickly?



PROPERTIES OF EQUALITY AND CONGRUENCE

Name: _____
Date: _____ Class: _____

Remember: In order to use the Transitive Property, the equivalent parts must each be an ENTIRE SIDE of the equation.



4

$$m\angle 1 + m\angle 2 = m\angle 5, \\ m\angle 2 = m\angle 3$$

$$m\angle 1 + m\angle 3 = m\angle 5$$

Transitive Property of Equality

$$m\angle 2 = m\angle 3$$

$$m\angle 2 + m\angle 7 = m\angle 3 + m\angle 7$$

Symmetric Property of Equality

Note: In a real proof, it would not really be very helpful to end up back where we started, but we are just practicing using the properties to justify each step.



5

$$\angle DGF \cong \angle FHK, \\ \angle FEC \cong \angle ABD, \\ \angle FHK \cong \angle FEC$$

$$\angle DGF \cong \angle ABD$$

Note: You may always go back to the beginning to find and use more of the given statements at any point



PROPERTIES OF EQUALITY AND CONGRUENCE

Answer Key

Name: _____
Date: _____ Class: _____

Complete the diagrams. The boxes should be filled in with equations and congruency statements. The arrows should be filled in with the properties needed to get from one step to another.

$$14m + 12 = 44 - 2m$$

Addition Property of Equality

$$14m + 2m + 12 = 44 - 2m + 2m$$

$$\text{or } 16m + 12 = 44$$

1

Subtraction Property of Equality

$$16m + 12 - 12 = 44 - 12$$

$$\text{Or } 16m = 32$$

Division Property of Equality

$$\frac{16m}{16} = \frac{32}{16} \text{ or } m = 2$$

$$\angle TUV \cong \angle VWX,$$

$$\angle TUV \cong \angle PQR$$

Transitive Property of Congruence

$$\angle VWX \cong \angle PQR$$

2

Symmetric Property of Congruence

$$\angle PQR \cong \angle VWX$$

$$m\angle 2 = m\angle 3,$$

$$m\angle 3 = m\angle 4$$

Addition Property of Equality

$$m\angle 3 + m\angle 5 = m\angle 4 + m\angle 5$$

3

Substitution Property of Equality

$$m\angle 2 + m\angle 5 = m\angle 4 + m\angle 5$$

Subtraction Property of Equality

$$m\angle 2 = m\angle 4$$

In diagram 3, which property could get from the first step to the last more quickly?

Transitive Property of Equality



PROPERTIES OF EQUALITY AND CONGRUENCE

Name: _____
Date: _____ Class: _____

Remember: In order to use the Transitive Property, the equivalent parts must each be an ENTIRE SIDE of the equation.

4

$$m\angle 1 + m\angle 2 = m\angle 5, \\ m\angle 2 = m\angle 3$$

Substitution Property of Equality

$$m\angle 1 + m\angle 3 = m\angle 5$$

Transitive Property of Equality

$$m\angle 1 + m\angle 2 = m\angle 1 + m\angle 3$$

Subtraction Property of Equality

$$m\angle 2 = m\angle 3$$

Addition Property of Equality

$$m\angle 2 + m\angle 7 = m\angle 3 + m\angle 7$$

Symmetric Property of Equality

$$m\angle 3 + m\angle 7 = m\angle 2 + m\angle 7$$

Note: In a real proof, it would not really be very helpful to end up back where we started, but we are just practicing using the properties to justify each step.

5

$$\angle DGF \cong \angle FHK, \\ \angle FEC \cong \angle ABD, \\ \angle FHK \cong \angle FEC$$

Transitive Property of Congruence

$$\angle DGF \cong \angle FEC$$

Transitive Property of Congruence

$$\angle DGF \cong \angle ABD$$

Note: You may always go back to the beginning to find and use more of the given statements at any point



PROPERTIES OF EQUALITY AND CONGRUENCE

WARM-UP / EXIT TICKET A

Name: _____

Date: _____ Class: _____

Identify the property.

1. Given that $m\angle BCA = m\angle DBC$ and $m\angle BCA = m\angle ADB$, we can conclude that $m\angle DBC = m\angle ADB$.
2. Given that $\overline{PQ} \cong \overline{TU}$, we can conclude that $\overline{TU} \cong \overline{PQ}$.
3. $\triangle ABC \cong \triangle ABC$
4. Given that $x = 3y$, we can conclude that $x - 13 = 3y - 13$.
5. Given that $m\angle 2 = m\angle 6$, we can conclude that $m\angle 2 + m\angle 4 = m\angle 6 + m\angle 4$.
6. Given that $\angle MND \cong \angle LPR$, we can rewrite the statement $\angle MND \cong \angle DEF$ as $\angle LPR \cong \angle DEF$.
7. Given that $c = a$, we can conclude that $7c = 7a$
8. Given that $\overline{GH} \cong \overline{VW}$ and $\overline{TU} \cong \overline{GH}$, we can conclude that $\overline{VW} \cong \overline{TU}$.
9. Given that $XY = MN$ and $XY + YZ = XZ$, we can state that $MN + YZ = XZ$.
10. If $4k = 18$ then $18 = 4k$.



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PROPERTIES OF EQUALITY AND CONGRUENCE

WARM-UP / EXIT TICKET B

Name: _____

Date: _____ Class: _____

Explain the difference between using Substitution and the Transitive Property.
Give one example in which either method can be used to reach the same conclusion and one example in which the Substitution cannot be accomplished using the Transitive Property.



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PROPERTIES OF EQUALITY AND CONGRUENCE

WARM-UP / EXIT TICKET A

Name: _____
Date: _____ Class: _____

Answer Key

Identify the property.

- Given that $m\angle BCA = m\angle DBC$ and $m\angle BCA = m\angle ADB$, we can conclude that $m\angle DBC = m\angle ADB$.
- Given that $\overline{PQ} \cong \overline{TU}$, we can conclude that $\overline{TU} \cong \overline{PQ}$.
- $\triangle ABC \cong \triangle ABC$
- Given that $x = 3y$, we can conclude that $x - 13 = 3y - 13$.
- Given that $m\angle 2 = m\angle 6$, we can conclude that $m\angle 2 + m\angle 4 = m\angle 6 + m\angle 4$.
- Given that $\angle MND \cong \angle LPR$, we can rewrite the statement $\angle MND \cong \angle DEF$ as $\angle LPR \cong \angle DEF$.
- Given that $c = a$, we can conclude that $7c = 7a$
- Given that $\overline{GH} \cong \overline{VW}$ and $\overline{TU} \cong \overline{GH}$, we can conclude that $\overline{VW} \cong \overline{TU}$.
- Given that $XY = MN$ and $XY + YZ = XZ$, we can state that $MN + YZ = XZ$.
- If $4k = 18$ then $18 = 4k$.

- Transitive Property of Equality
- Symmetric Property of Congruence
- Reflexive Property of Congruence
- Subtraction Property of Equality
- Addition Property of Equality
- Substitution Property of Congruence
- Multiplication Property of Equality
- Transitive Property of Congruence
- Substitution Property of Equality
- Symmetric Property of Equality



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PROPERTIES OF EQUALITY AND CONGRUENCE

WARM-UP / EXIT TICKET B

Name: _____
Date: _____ Class: _____

Explain the difference between using Substitution and the Transitive Property.
Give one example in which either method can be used to reach the same conclusion and one example in which the Substitution cannot be accomplished using the Transitive Property.

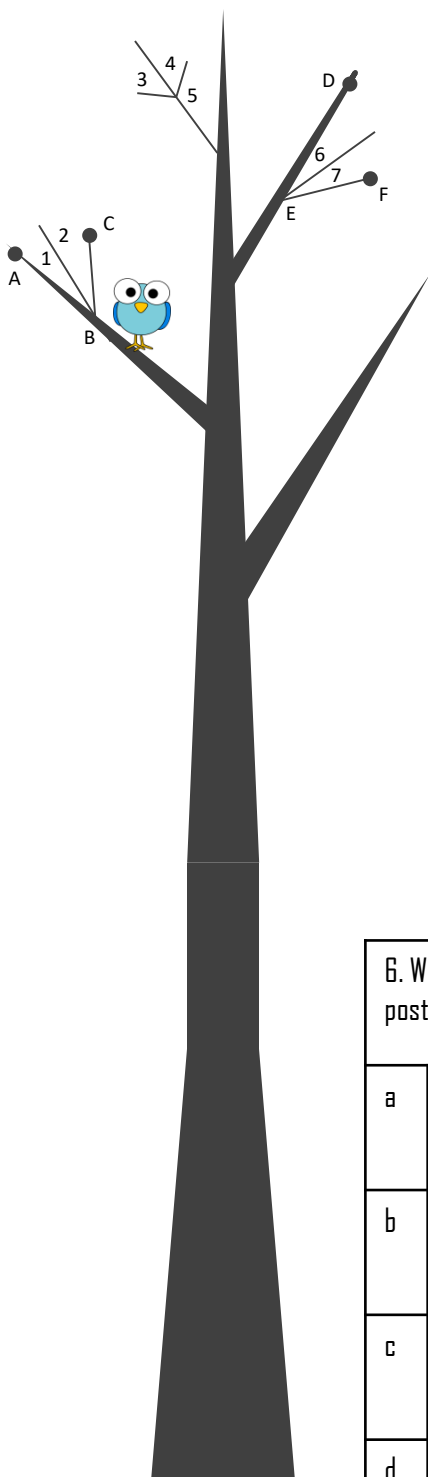


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SEGMENT ADDITION POSTULATE AND ANGLE ADDITION POSTULATE

Name: _____

Date: _____ Class: _____



- Given that the distance from the ground to the lowest branch is 39 meters and the distance from the ground to the branch with the bird is 46 meters, find the distance between the lowest branch and the branch with the bird.
- Given that the distance from the branch with the bird to the top of the tree is 15 meters, find the height of the tree.
- Given that the measure of angle 1 is 28 degrees and the measure of angle 2 is 55 degrees, find the measure of angle ABC.
- Given that the measure of angle DEF is 62 degrees and angle 6 is congruent to angle 7, find the measure of angle 6.
- Given that the measure of angle 3 is 23 degrees, the measure of angle 5 is 134 degrees, and the measure of the large angle made up of all three angles 3, 4, and 5 is 221 degrees, Find the measure of angle 4.

6. Write an equation that can be used to find the requested measure and identify which postulate you used.

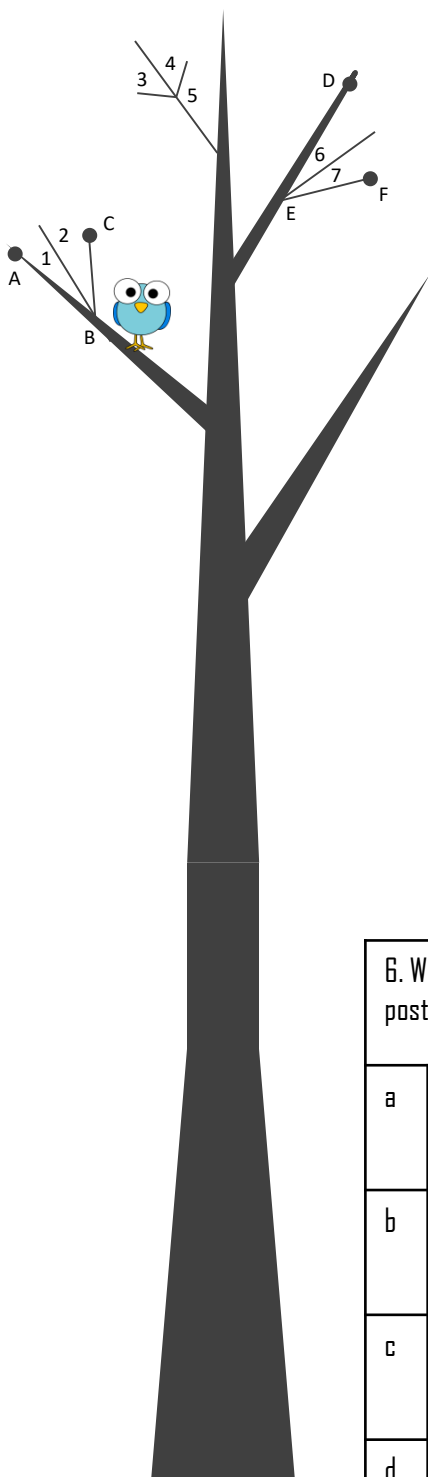
a		Find $m\angle MOP$.	Equation: _____	Postulate: _____
b		Find XZ.	Equation: _____	Postulate: _____
c		Find $m\angle AED$.	Equation: _____	Postulate: _____
d		Find SW.	Equation: _____	Postulate: _____



SEGMENT ADDITION POSTULATE AND ANGLE ADDITION POSTULATE

Name: _____
Date: _____ Class: _____

Answer Key



1. Given that the distance from the ground to the lowest branch is 39 meters and the distance from the ground to the branch with the bird is 46 meters, find the distance between the lowest branch and the branch with the bird.

7 meters

2. Given that the distance from the branch with the bird to the top of the tree is 15 meters, find the height of the tree.

61 meters

3. Given that the measure of angle 1 is 28° and the measure of angle 2 is 55° , find the measure of angle ABC.

83°

4. Given that the measure of angle DEF is 62 degrees and angle 6 is congruent to angle 7, find the measure of angle 6.

31°

5. Given that the measure of angle 3 is 23 degrees, the measure of angle 5 is 134 degrees, and the measure of the large angle made up of all three angles 3, 4, and 5 is 221 degrees, Find the measure of angle 4.

64°

6. Write an equation that can be used to find the requested measure and identify which postulate you used.

a		Find $m\angle MOP$.	Equation: $m\angle MON + m\angle NOP = m\angle MOP$	Postulate: Angle Addition Postulate
b		Find XZ.	Equation: $XY + YZ = XZ$	Postulate: Segment Addition Postulate
c		Find $m\angle AED$.	Equation: $m\angle AEB + m\angle BEC + m\angle CED = m\angle AED$	Postulate: Angle Addition Postulate
d		Find SW.	Equation: $ST + TU + UV + VW = SW$	Postulate: Segment Addition Postulate



TWO- COLUMN STRUCTURE FOR PROOFS: ALGEBRA PROOFS

Name: _____
Date: _____ Class: _____

Fill in the missing justifications in each proof.

1. Given: $a = 2b + 6$
 $a = 9b - 8$

Prove: $b = 2$

2. Given: $\frac{x}{2} = y + 5$
 $y = 3x$

Prove: $x = -2$



	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1	$a = 2b + 6$		1	$\frac{x}{2} = y + 5$	
2	$a = 9b - 8$		2	$y = 3x$	
3	$2b + 6 = 9b - 8$		3	$\frac{x}{2} = 3x + 5$	
4	$6 = 7b - 8$		4	$x = 6x + 10$	
5	$14 = 7b$		5	$-5x = 10$	
6	$2 = b$		6	$x = -2$	
7	$b = 2$		7		
8			8		

3. Given: $3w = t$
 $2r = s$
 $t = 2r$

Prove: $3w = s$

4. Given: $3b + d = f$
 $d = 2b$
 $f = g$

Prove: $g = 5b$

	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1	$3w = t$		1	$3b + d = f$	
2	$2r = s$		2	$d = 2b$	
3	$t = 2r$		3	$f = g$	
4	$3w = 2r$		4	$3b + 2b = f$	
5	$3w = s$		5	$5b = f$	(simplified step 5)
6			6	$5b = g$	
7			7	$g = 5b$	
8			8		



TWO-COLUMN STRUCTURE FOR PROOFS: ALGEBRA PROOFS

Name: _____
Date: _____ Class: _____

Complete each proof.

5. Given: $5c + 1 = 2c - 8$

Prove: $c = -3$

6. Given: $m = n + 5$
 $2m = n$

Prove: $m = -5$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

7. Given: $g = 2h$
 $g + h = k$
 $k = m$

Prove: $m = 3h$



8. Given: $ab = 8a$
 $b + d = 20$

Prove: $d = 12$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		



TWO- COLUMN STRUCTURE FOR PROOFS: ALGEBRA PROOFS

Name: _____
Date: _____ Class: _____

Answer Key

Fill in the missing justifications in each proof.

1. Given: $a = 2b + 6$
 $a = 9b - 8$

Prove: $b = 2$

2. Given: $\frac{x}{2} = y + 5$
 $y = 3x$

Prove: $x = -2$



	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1	$a = 2b + 6$	Given	1	$\frac{x}{2} = y + 5$	Given
2	$a = 9b - 8$	Given	2	$y = 3x$	Given
3	$2b + 6 = 9b - 8$	Transitive Prop. (1, 2)	3	$\frac{x}{2} = 3x + 5$	Substitution (1, 2)
4	$6 = 7b - 8$	Subtraction Prop. of Eq.	4	$x = 6x + 10$	Multiplication Prop. of Eq.
5	$14 = 7b$	Addition Prop. of Eq.	5	$-5x = 10$	Subtraction Prop. Of Eq.
6	$2 = b$	Division Prop. Of Eq.	6	$x = -2$	Division Prop. Of Eq.
7	$b = 2$	Symmetric Prop.	7		
8			8		

3. Given: $3w = t$
 $2r = s$
 $t = 2r$

Prove: $3w = s$

4. Given: $3b + d = f$
 $d = 2b$
 $f = g$

Prove: $g = 5b$

	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1	$3w = t$	Given	1	$3b + d = f$	Given
2	$2r = s$	Given	2	$d = 2b$	Given
3	$t = 2r$	Given	3	$f = g$	Given
4	$3w = 2r$	Transitive Prop. (1, 3)	4	$3b + 2b = f$	Substitution (1, 2)
5	$3w = s$	Transitive Prop. (2, 4)	5	$5b = f$	(simplified step 5)
6			6	$5b = g$	Transitive Prop. (3, 5)
7			7	$g = 5b$	Symmetric Prop.
8			8		



TWO-COLUMN STRUCTURE FOR PROOFS: ALGEBRA PROOFS

Name: _____
Date: _____ Class: _____

Answer Key

Complete each proof.

5. Given: $5c + 1 = 2c - 8$

Prove: $c = -3$

6. Given: $m = n + 5$
 $2m = n$

Prove: $m = -5$

	STATEMENT	JUSTIFICATION
1	$5c + 1 = 2c - 8$	Given
2	$3c + 1 = -8$	Subtraction Prop. of Eq.
3	$3c = -9$	Subtraction Prop. of Eq.
4	$c = -3$	Division Prop. of Eq.
5		
6		
7		
8		

	STATEMENT	JUSTIFICATION
1	$m = n + 5$	Given
2	$2m = n$	Given
3	$m = 2m + 5$	Substitution (1, 2)
4	$-m = 5$	Subtraction Prop. of Eq.
5	$m = -5$	Division Prop. of Eq.
6		
7		
8		

7. Given: $g = 2h$
 $g + h = k$
 $k = m$

Prove: $m = 3h$



8. Given: $ab = 8a$
 $b + d = 20$

Prove: $d = 12$

	STATEMENT	JUSTIFICATION
1	$g = 2h$	Given
2	$g + h = k$	Given
3	$k = m$	given
4	$2h + h = k$	Substitution (1, 2)
5	$3h = k$	(Simplified line 4)
6	$3h = m$	Transitive Prop. (3, 5)
7	$m = 3h$	Symmetric Prop.
8		

	STATEMENT	JUSTIFICATION
1	$ab = 8a$	Given
2	$b + d = 20$	Given
3	$b = 8$	Division Prop. of Eq.
4	$8 + d = 20$	Substitution (2, 3)
5	$d = 12$	Subtraction Prop. of Eq.
6		
7		
8		



ALGEBRA PROOFS

WARM-UP / EXIT TICKET A

Name: _____
Date: _____ Class: _____

Given: $w + 4 = 7w - 2$
 $v = w$

Prove: $v = 1$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

ALGEBRA PROOFS

WARM-UP / EXIT TICKET B

Name: _____
Date: _____ Class: _____

Given: $c + d = f$
 $f = k$
 $c = d$

Prove: $k = 2c$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		



ALGEBRA PROOFS

WARM-UP / EXIT TICKET A

Name: _____
Date: _____ Class: _____

Answer Key

Given: $w + 4 = 7w - 2$
 $v = w$

Prove: $v = 1$

	STATEMENT	JUSTIFICATION
1	$w + 4 = 7w - 2$	Given
2	$v = w$	Given
3	$4 = 6w - 2$	Subtraction Prop. of Eq.
4	$6 = 6w$	Addition Prop. of Eq.
5	$1 = w$	Division Prop. of Eq.
6	$1 = v$	Substitution (2, 5)
7	$v = 1$	Symmetric Prop. of Eq.
8		

ALGEBRA PROOFS

WARM-UP / EXIT TICKET B

Name: _____
Date: _____ Class: _____

Answer Key

Given: $c + d = f$
 $f = k$
 $c = d$

Prove: $k = 2c$

	STATEMENT	JUSTIFICATION
1	$c + d = f$	Given
2	$f = k$	Given
3	$c = d$	Given
4	$c + c = f$	Substitution (1, 3)
5	$2c = f$	(simplified line 4)
6	$2c = k$	Transitive Prop. (2, 5)
7	$k = 2c$	Symmetric Prop. Of Eq.
8		



GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES

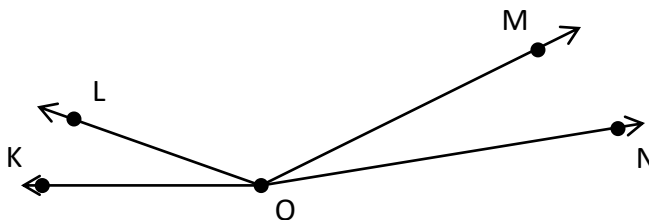
Name: _____

Date: _____ Class: _____

1

Given: $m\angle KOL = m\angle MON$

Prove: $m\angle KOM = m\angle LON$

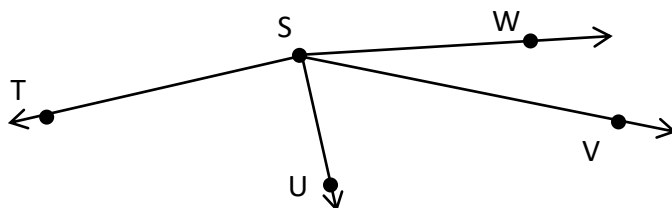


	STATEMENT	JUSTIFICATION
1		
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10		

2

Given: $\overrightarrow{SU}$ bisects $\angle TSW$

Prove: $m\angle USV + m\angle VSW = m\angle TSU$



	STATEMENT	JUSTIFICATION
1		
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9		
10		



GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES

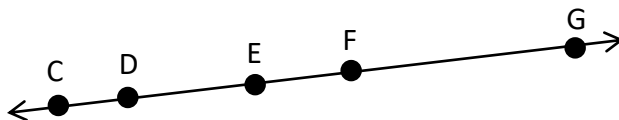
Name: _____

Date: _____ Class: _____

3

Given: $\overline{CD} \cong \overline{EF}$
 $\overline{DE} \cong \overline{FG}$

Prove: $\overline{CE} \cong \overline{EG}$

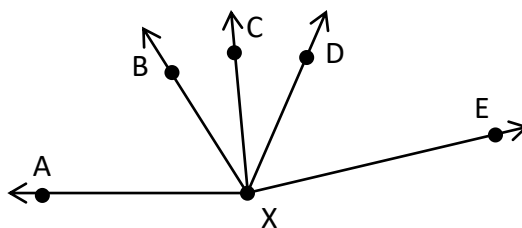


	STATEMENT	JUSTIFICATION
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4

Given: $\overrightarrow{XB}$ bisects $\angle AXE$
 $\overrightarrow{XD}$ bisects $\angle CXE$

Prove: $m\angle DXE + m\angle AXB = m\angle BXD$



	STATEMENT	JUSTIFICATION
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GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES

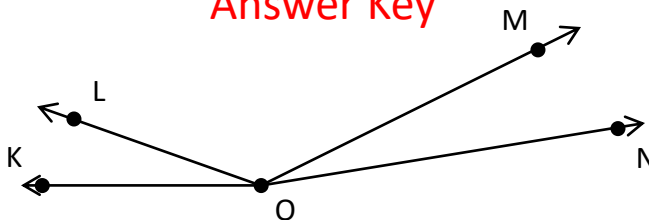
Name: _____
Date: _____ Class: _____

1

Given: $m\angle KOL = m\angle MON$

Prove: $m\angle KOM = m\angle LON$

Answer Key

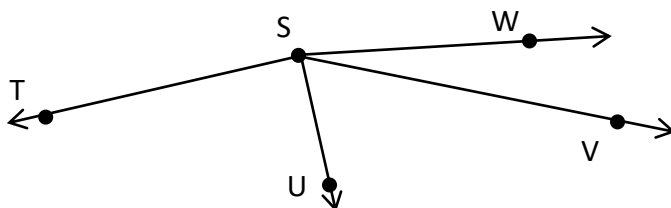


	STATEMENT	JUSTIFICATION
1	$m\angle KOL = m\angle MON$	Given
2	$m\angle KOL + m\angle LOM = m\angle KOM$	Angle Addition Postulate
3	$m\angle MON + m\angle LOM = m\angle LON$	Angle Addition Postulate
4	$m\angle KOL + m\angle LOM = m\angle LON$	Substitution (1, 3)
5	$m\angle KOM = m\angle LON$	Transitive Prop. (2, 4)
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2

Given: $\overrightarrow{SU}$ bisects $\angle TSW$

Prove: $m\angle USV + m\angle VSW = m\angle TSU$



	STATEMENT	JUSTIFICATION
1	$\overrightarrow{SU}$ bisects $\angle TSW$	Given
2	$m\angle TSU = m\angle USW$	Defn. bisector
3	$m\angle USV + m\angle VSW = m\angle USW$	Angle Addition Postulate
4	$m\angle USV + m\angle VSW = m\angle TSU$	Substitution (2, 3)
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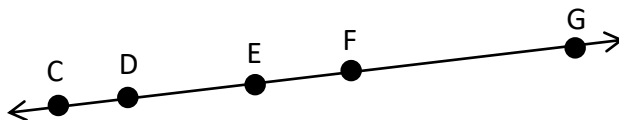
GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES

Name: _____
Date: _____ Class: _____

3

Given: $\overline{CD} \cong \overline{EF}$
 $\overline{DE} \cong \overline{FG}$

Prove: $\overline{CE} \cong \overline{EG}$

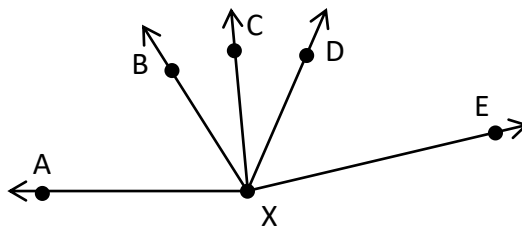


	STATEMENT	JUSTIFICATION
1	$\overline{CD} \cong \overline{EF}$	Given
2	$\overline{DE} \cong \overline{FG}$	Given
3	$CD = EF$	Defn. Congruent
4	$DE = FG$	Defn. congruent
5	$CD + DE = CE$	Segment Addition Postulate
6	$EF + FG = EG$	Segment Addition Postulate
7	$CD + FG = EG$	Substitution (3, 6)
8	$CD + DE = EG$	Substitution (4, 7)
9	$CE = EG$	Transitive Prop. (5, 8)
10	$\overline{CE} \cong \overline{EG}$	Defn. congruent

4

Given: $\overrightarrow{XB}$ bisects $\angle AXC$
 $\overrightarrow{XD}$ bisects $\angle CXE$

Prove: $m\angle DXE + m\angle AXB = m\angle BXD$



	STATEMENT	JUSTIFICATION
1	$\overrightarrow{XB}$ bisects $\angle AXC$	Given
2	$\overrightarrow{XD}$ bisects $\angle CXE$	Given
3	$m\angle AXB = m\angle BXC$	Defn. bisector
4	$m\angle CXD = m\angle DXE$	Defn. bisector
5	$m\angle CXD + m\angle BXC = m\angle BXD$	Angle Addition Postulate
6	$m\angle DXE + m\angle BXC = m\angle BXD$	Substitution (4, 5)
7	$m\angle DXE + m\angle AXB = m\angle BXD$	Substitution (3, 6)
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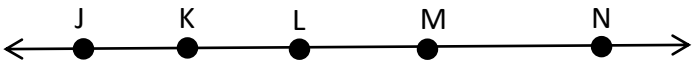
GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES WARM-UP / EXIT TICKET A

Name: _____

Date: _____ Class: _____

Given: $\overline{JM} \cong \overline{KN}$

Prove: $\overline{JK} \cong \overline{MN}$



	STATEMENT	JUSTIFICATION
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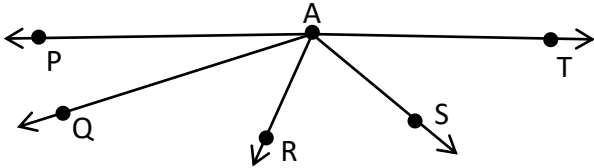
GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES WARM-UP / EXIT TICKET B

Name: _____

Date: _____ Class: _____

Given: $\angle PAQ \cong \angle RAS$, $\angle RAS \cong \angle SAT$

Prove: $\angle PAS \cong \angle QAT$



	STATEMENT	JUSTIFICATION
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GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES WARM-UP / EXIT TICKET A

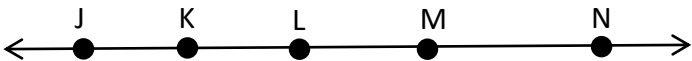
Name: _____

Date: _____ Class: _____

Answer Key

Given: $\overline{JM} \cong \overline{KN}$

Prove: $\overline{JK} \cong \overline{MN}$



	STATEMENT	JUSTIFICATION
1	$\overline{JM} \cong \overline{KN}$	Given
2	$JM = KN$	Defn. congruent
3	$JK + KL + LM = JM$	Segment Addition Postulate
4	$KL + LM + MN = KN$	Segment Addition Postulate
5	$KL + LM + MN = JM$	Substitution (2, 4)
6	$JK + KL + LM = KL + LM + MN$	Transitive Prop. (3, 5)
7	$JK = MN$	Subtraction Property of Eq.
8	$\overline{JK} \cong \overline{MN}$	Defn. congruent
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GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES WARM-UP / EXIT TICKET B

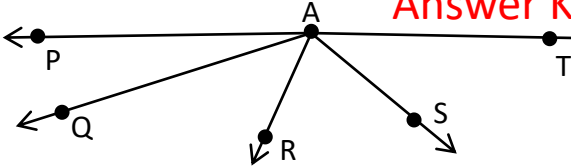
Name: _____

Date: _____ Class: _____

Answer Key

Given: $\angle PAQ \cong \angle RAS, \angle RAS \cong \angle SAT$

Prove: $\angle PAS \cong \angle QAT$



	STATEMENT	JUSTIFICATION
1	$\angle PAQ \cong \angle RAS$	Given
2	$\angle RAS \cong \angle SAT$	Given
3	$m\angle PAQ = m\angle RAS$	Defn. congruent
4	$m\angle RAS = m\angle SAT$	Defn. congruent
5	$m\angle PAQ = m\angle SAT$	Transitive Prop. (3, 4)
6	$m\angle PAQ + m\angle QAR + m\angle RAS = m\angle PAS$	Angle Addition Postulate
7	$m\angle QAR + m\angle RAS + m\angle SAT = m\angle QAT$	Angle Addition Postulate
8	$m\angle QAR + m\angle RAS + m\angle PAQ = m\angle QAT$	Substitution (5, 7)
9	$m\angle PAS = m\angle QAT$	Transitive Prop. (6, 8)
10	$\angle PAS \cong \angle QAT$	Defn. congruent

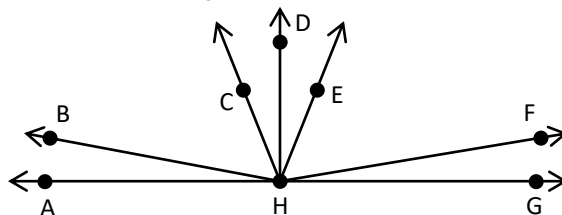


Name: _____

PROOF PUZZLE: CUT OUT STATEMENTS AND JUSTIFICATIONS AND SEE IF YOU CAN ARRANGE THEM TO DEVELOP A TWO COLUMN PROOF. YOU MIGHT NOT USE ALL OF THE PIECES. THERE MAY BE MORE THAN ONE CORRECT FINISHED PROOF.

Given: $m\angle AHB = m\angle FHG$, $m\angle CHD = m\angle DHE$, $m\angle BHD = m\angle DHF$

Prove: $m\angle CHF = m\angle BHE$



	STATEMENT	JUSTIFICATION
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STATEMENTS AND JUSTIFICATIONS TO CUT OUT FOR PROOF PUZZLE

Transitive Property (,)

$$m\angle DHE + m\angle EHF = m\angle DHF$$

Given

Angle Addition Postulate

$$m\angle AHB = m\angle FHG$$

$$m\angle CHD + m\angle DHE + m\angle EHF = m\angle CHF$$

Angle Addition Postulate

Transitive Property (,)

$$m\angle BHC + m\angle CHD = m\angle CHD + m\angle EHF$$

Substitution (,)

$$m\angle BHD = m\angle DHF$$

Given

Subtraction Property of Equality

$$m\angle AHB = m\angle EHF$$

Substitution (,)

$$m\angle CHD + m\angle EHF = m\angle DHF$$

Given

$$m\angle BHC + m\angle CHD = m\angle BHD$$

Substitution (,)

$$m\angle CHF = m\angle BHE$$

$$m\angle CHD + m\angle EHF = m\angle BHD$$

Transitive Property (,)

Angle Addition Postulate

$$m\angle EHF + m\angle CHD + m\angle DHE = m\angle BHE$$

$$m\angle CHD = m\angle DHE$$

$$m\angle BHC = m\angle EHF$$

Substitution (,)

Substitution (,)

Angle Addition Postulate

$$m\angle BHC + m\angle CHD + m\angle DHE = m\angle BHE$$

Given

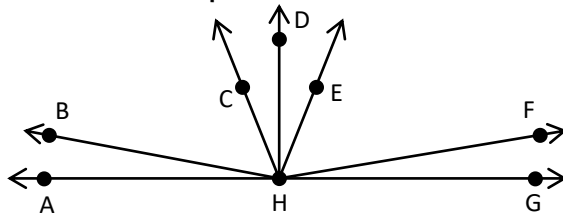
Addition Property of Equality



PROOF PUZZLE: CUT OUT STATEMENTS AND JUSTIFICATIONS AND SEE IF YOU CAN ARRANGE THEM TO DEVELOP A TWO COLUMN PROOF. YOU MIGHT NOT USE ALL OF THE PIECES. THERE MAY BE MORE THAN ONE CORRECT FINISHED PROOF.

Given: $m\angle AHB = m\angle FHG$, $m\angle CHD = m\angle DHE$, $m\angle BHD = m\angle DHF$

Prove: $m\angle CHF = m\angle BHE$



	STATEMENT	JUSTIFICATION
1	$m\angle AHB = m\angle FHG$	Given
2	$m\angle CHD = m\angle DHE$	Given
3	$m\angle BHD = m\angle DHF$	Given
4	$m\angle BHC + m\angle CHD = m\angle BHD$	Angle Addition Postulate
5	$m\angle DHE + m\angle EHF = m\angle DHF$	Angle Addition Postulate
6	$m\angle CHD + m\angle EHF = m\angle DHF$	Substitution (2, 5)
7	$m\angle CHD + m\angle EHF = m\angle BHD$	Substitution (3, 6)
8	$m\angle BHC + m\angle CHD = m\angle CHD + m\angle EHF$	Transitive Property (4, 7)
9	$m\angle BHC = m\angle EHF$	Subtraction Property of Equality
10	$m\angle CHD + m\angle DHE + m\angle EHF = m\angle CHF$	Angle Addition Postulate
11	$m\angle BHC + m\angle CHD + m\angle DHE = m\angle BHE$	Angle Addition Postulate
12	$m\angle EHF + m\angle CHD + m\angle DHE = m\angle BHE$	Substitution (9, 11)
13	$m\angle CHF = m\angle BHE$	Transitive Property (10, 12)

QUIZ: WRITING TWO-COLUMN PROOFS

Complete each proof.

Name: _____

Date: _____ Class: _____

1. Given: $a + c = 3f$
 $c = d$
 $a = 2d$

Prove: $d = f$

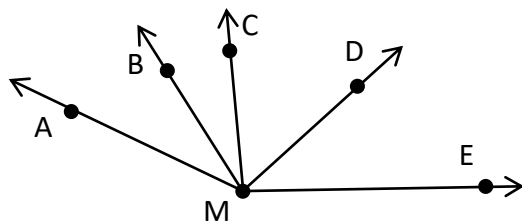
2. Given: $x = y$
 $z = y$
 $w = x$
 $w = t$

Prove: $t = z$

	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1			1		
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8			8		

3. Given: $\overrightarrow{MD}$ bisects $\angle CME$, $\angle AMB \cong \angle BMC$

Prove: $m\angle BMD = m\angle AMB + m\angle DME$

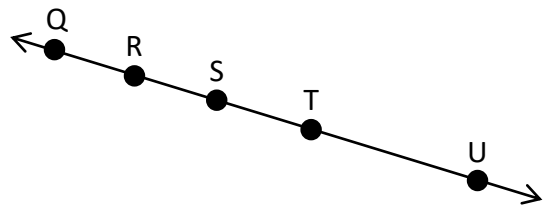


	STATEMENT	JUSTIFICATION
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4. Given: $QT = SU$

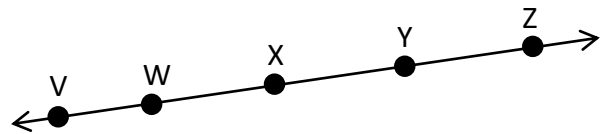
Prove: $QS = TU$



	STATEMENT	JUSTIFICATION
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5. Given: X is the midpoint of $\overline{VZ}$ and of $\overline{WY}$
 W is the midpoint of $\overline{VX}$
 Y is the midpoint of $\overline{XZ}$

Prove: $VY = WZ$



	STATEMENT	JUSTIFICATION
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QUIZ: WRITING TWO-COLUMN PROOFS

Complete each proof.

Name: _____

Date: _____ Class: _____

Answer Key

1. Given: $a + c = 3f$
 $c = d$
 $a = 2d$

Prove: $d = f$

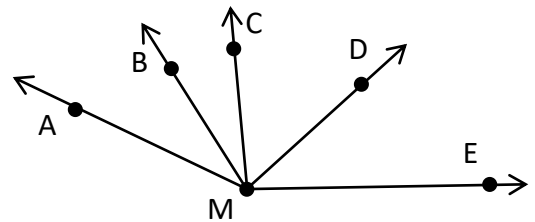
2. Given: $x = y$
 $z = y$
 $w = x$
 $w = t$

Prove: $t = z$

	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1	$a + c = 3f$	Given	1	$x = y$	Given
2	$c = d$	Given	2	$z = y$	Given
3	$a = 2d$	Given	3	$w = x$	Given
4	$2d + c = 3f$	Substitution (1, 3)	4	$w = t$	Given
5	$2d + d = 3f$	Substitution (2, 4)	5	$t = x$	Transitive Prop. (3, 4)
6	$3d = 3f$	Simplified line 5	6	$t = y$	Transitive Prop. (1, 5)
7	$d = f$	Division Prop. of Eq.	7	$t = z$	Transitive Prop. (2, 6)
8			8		

3. Given: $\overrightarrow{MD}$ bisects $\angle CME$, $\angle AMB \cong \angle BMC$

Prove: $m\angle BMD = m\angle AMB + m\angle DME$

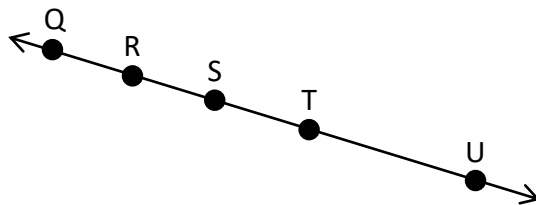


	STATEMENT	JUSTIFICATION
1	$\overrightarrow{MD}$ bisects $\angle CME$	Given
2	$\angle AMB \cong \angle BMC$	Given
3	$m\angle CMD = m\angle DME$	Defn. bisector
4	$m\angle AMB = m\angle BMC$	Defn. congruent
5	$m\angle BMC + m\angle CMD = m\angle BMD$	Angle Addition Postulate
6	$m\angle AMB + m\angle CMD = m\angle BMD$	Substitution (4, 5)
7	$m\angle AMB + m\angle DME = m\angle BMD$	Substitution (3, 6)
8	$m\angle BMD = m\angle AMB + m\angle DME$	Symmetric Prop. of Eq.
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4. Given: $QT = SU$

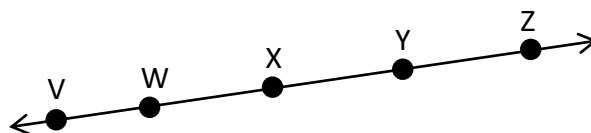
Prove: $QS = TU$



	STATEMENT	JUSTIFICATION
1	$QT = SU$	Given
2	$QS + ST = QT$	Segment Addition Postulate
3	$ST + TU = SU$	Segment Addition Postulate
4	$ST + TU = QT$	Substitution (1, 3)
5	$QS + ST = ST + TU$	Transitive Prop. (2, 4)
6	$QS = TU$	Subtraction Prop. of Eq.
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5. Given: X is the midpoint of $\overline{VZ}$ and of $\overline{WY}$
 W is the midpoint of $\overline{VX}$
 Y is the midpoint of $\overline{XZ}$

Prove: $VY = WZ$



	STATEMENT	JUSTIFICATION
1	X is the midpoint of $\overline{VZ}$ and of $\overline{WY}$	Given
2	W is the midpoint of $\overline{VX}$	Given
3	Y is the midpoint of $\overline{XZ}$	Given
4	$VX = XZ$	Defn. midpoint
5	$WX = XY$	Defn. midpoint
6	$VW = WX$	Defn. midpoint
7	$XY = YZ$	Defn. midpoint
8	$VW + WX + XY = VY$	Segment Addition Postulate
9	$WX + XY + YZ = WZ$	Segment Addition Postulate
10	$VW = XY$	Transitive Prop. (5, 6)
11	$VW = YZ$	Transitive Prop (7, 10)
12	$WX + XY + VW = WZ$	Substitution (9, 11)
13	$VY = WZ$	Transitive Prop. (8, 12)

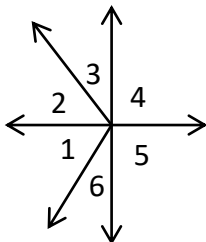


PROOFS USING SPECIAL ANGLE PAIRS

Name: _____
Date: _____ Class: _____

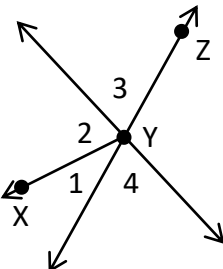
1. Given: $\angle 4$ is a right angle,
 $\angle 1 \cong \angle 2$

Prove: $\angle 2$ and $\angle 6$ are
complementary.



2. Given: $\angle 2$ is a right
angle,
 $m\angle 4 = 60^\circ$

Prove: $m\angle XYZ = 150^\circ$

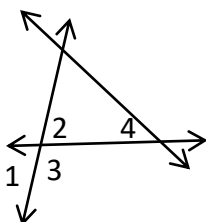


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	STATEMENT	JUSTIFICATION
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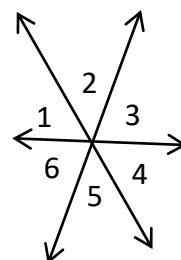
3. Given: $\angle 3$ and $\angle 4$ are
supplements.

Prove: $\angle 2 \cong \angle 4$



4. Given: $\angle 1 \cong \angle 2$

Prove: $\angle 5 \cong \angle 4$



	STATEMENT	JUSTIFICATION
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	STATEMENT	JUSTIFICATION
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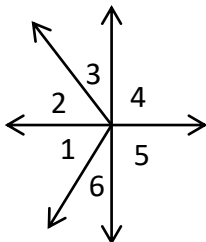
PROOFS USING SPECIAL ANGLE PAIRS

Name: _____
Date: _____ Class: _____

Answer Key

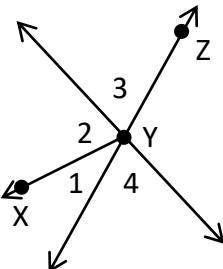
1. Given: $\angle 4$ is a right angle,
 $\angle 1 \cong \angle 2$

Prove: $\angle 2$ and $\angle 6$ are complementary.



2. Given: $\angle 2$ is a right angle,
 $m\angle 4 = 60^\circ$

Prove: $m\angle XYZ = 150^\circ$

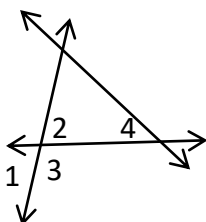


	STATEMENT	JUSTIFICATION
1	$\angle 4$ is a right angle	Given
2	$\angle 1 \cong \angle 2$	Given
3	$m\angle 4 = 90$	Defn. right angle
4	$m\angle 1 = m\angle 2$	Defn. congruent
5	$m\angle 1 + m\angle 6 = m\angle 4$	Vertical Angles Theorem
6	$m\angle 1 + m\angle 6 = 90$	Substitution (3, 5)
7	$m\angle 2 + m\angle 6 = 90$	Substitution (4, 6)
8	$\angle 2$ and $\angle 6$ are complementary	Defn. complementary

	STATEMENT	JUSTIFICATION
1	$\angle 2$ is a right angle	Given
2	$m\angle 4 = 60$	Given
3	$m\angle 2 = 90$	Defn. right angle
4	$m\angle 2 + m\angle 3 = m\angle XYZ$	Angle Addition Post.
5	$m\angle 3 = m\angle 4$	Vertical Angles Theorem
6	$m\angle 2 + m\angle 4 = m\angle XYZ$	Substitution (4, 5)
7	$90 + 60 = m\angle XYZ$	Substitution (2, 3, 6)
8	$150 = m\angle XYZ$	Simplified line 7
9	$m\angle XYZ = 150$	Symmetric Prop. Of Eq.

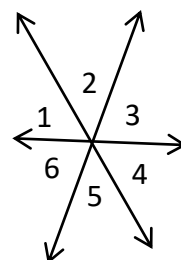
3. Given: $\angle 3$ and $\angle 4$ are supplements.

Prove: $\angle 2 \cong \angle 4$



4. Given: $\angle 1 \cong \angle 2$

Prove: $\angle 5 \cong \angle 4$



	STATEMENT	JUSTIFICATION
1	$\angle 3$ and $\angle 4$ are suppl.	Given
2	$m\angle 3 + m\angle 4 = 180$	Defn. suppl.
3	$m\angle 2 + m\angle 3 = 180$	Linear Pair Theorem
4	$m\angle 2 + m\angle 3 = m\angle 3 + m\angle 4$	Transitive Prop. (2, 3)
5	$m\angle 2 = m\angle 4$	Subtraction Prop. Of Eq.
6	$\angle 2 \cong \angle 4$	Defn. congruent
7		
8		

	STATEMENT	JUSTIFICATION
1	$\angle 1 \cong \angle 2$	Given
2	$\angle 2 \cong \angle 5$	Vertical Angles Theorem
3	$\angle 1 \cong \angle 4$	Vertical Angles Theorem
4	$\angle 1 \cong \angle 5$	Transitive Prop. (1, 2)
5	$\angle 5 \cong \angle 4$	Transitive Prop. (3, 4)
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7		
8		



GEOMETRY PROOFS USING SPECIAL ANGLE
PAIRS WARM-UP / EXIT TICKET A

Name: _____

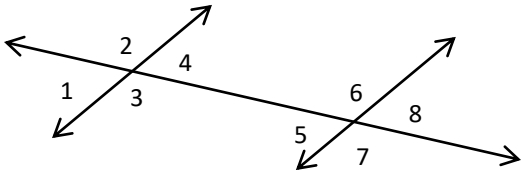
Date: _____ Class: _____

Given:

$\angle 4 \cong \angle 8$

Prove:

$\angle 4 \cong \angle 5$



	STATEMENT	JUSTIFICATION
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GEOMETRY PROOFS USING SPECIAL ANGLE
PAIRS WARM-UP / EXIT TICKET B

Name: _____

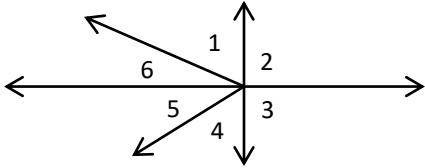
Date: _____ Class: _____

Given:

$\angle 2$ is a right angle, $\angle 4 \cong \angle 6$

Prove:

$\angle 5$ and $\angle 6$ are complementary.



	STATEMENT	JUSTIFICATION
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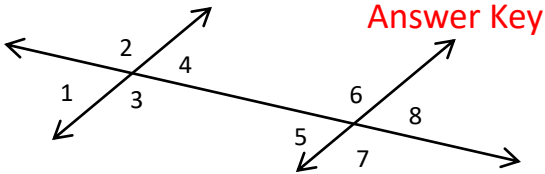
GEOMETRY PROOFS USING SPECIAL ANGLE
PAIRS WARM-UP / EXIT TICKET A

Name: _____

Date: _____ Class: _____

Given: $\angle 4 \cong \angle 8$

Prove: $\angle 4 \cong \angle 5$



	STATEMENT	JUSTIFICATION
1	$\angle 4 \cong \angle 8$	Given
2	$\angle 5 \cong \angle 8$	Vertical Angles Theorem
3	$\angle 4 \cong \angle 5$	Transitive Prop. (1, 2)
4		
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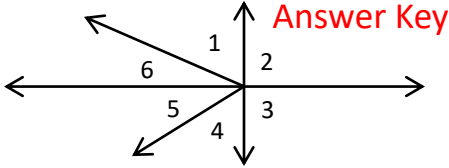
GEOMETRY PROOFS USING SPECIAL ANGLE
PAIRS WARM-UP / EXIT TICKET B

Name: _____

Date: _____ Class: _____

Given: $\angle 2$ is a right angle, $\angle 4 \cong \angle 6$

Prove: $\angle 5$ and $\angle 6$ are complementary.



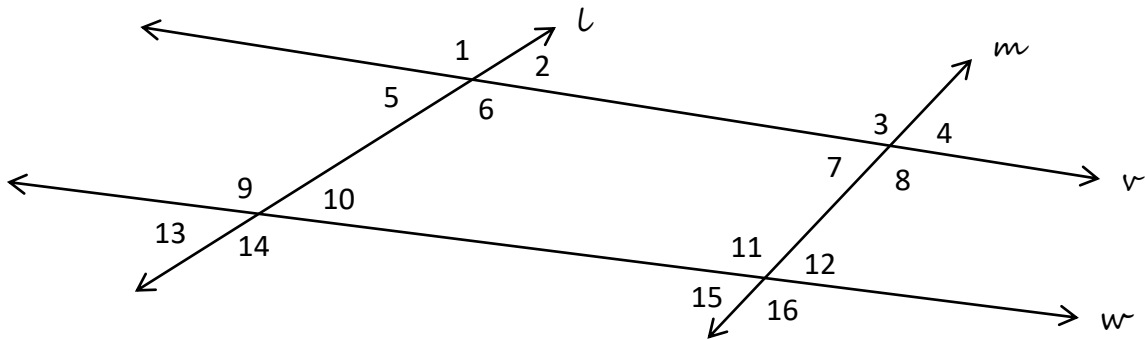
	STATEMENT	JUSTIFICATION
1	$\angle 2$ is a right angle	Given
2	$\angle 4 \cong \angle 6$	Given
3	$m\angle 2 = 90$	Defn. right angle
4	$m\angle 4 = m\angle 6$	Defn. congruent
5	$m\angle 4 + m\angle 5 = m\angle 2$	Vertical Angles Theorem
6	$m\angle 4 + m\angle 5 = 90$	Substitution (4, 5))
7	$m\angle 6 + m\angle 5 = 90$	Substitution (4, 6)
8	$\angle 5$ and $\angle 6$ are complementary	Defn. complementary
9		
10		



PROOFS WITH SPECIAL ANGLES ALONG TRANSVERSALS

Name: _____

Date: _____ Class: _____



1. Given: $\angle 2 \cong \angle 7$
Prove: $\angle 14 \cong \angle 16$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

2. Given: $v \parallel w$, $\angle 7 \cong \angle 11$
Prove: $m\angle 16 = 90^\circ$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		
9		

3. Given: $l \parallel m$, $v \parallel w$
Prove: $\angle 1 \cong \angle 16$

	STATEMENT	JUSTIFICATION
1		
2		
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5		
6		
7		
8		

4. Given: $\angle 13 \cong \angle 4$, $l \parallel m$
Prove: $v \parallel w$

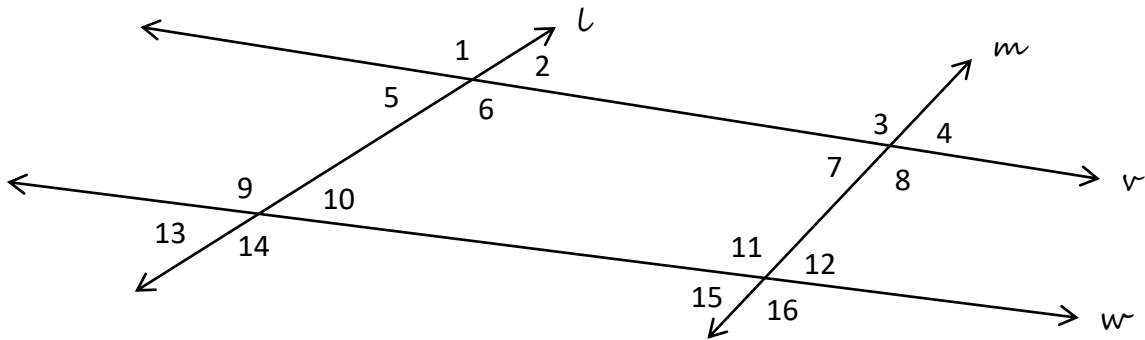
	STATEMENT	JUSTIFICATION
1		
2		
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7		
8		



PROOFS WITH SPECIAL ANGLES ALONG TRANSVERSALS

Name: _____

Date: _____ Class: _____



1. Given: $\angle 2 \cong \angle 7$
Prove: $\angle 14 \cong \angle 16$

	STATEMENT	JUSTIFICATION
1	$\angle 2 \cong \angle 7$	Given
2	$l \parallel m$	Converse of Alt. Int. Angles Thm.
3	$\angle 4 \cong \angle 6$	Corresp. Angles Post.
4		
5		
6		
7		
8		

2. Given: $v \parallel w$, $\angle 7 \cong \angle 11$
Prove: $m\angle 16 = 90^\circ$

	STATEMENT	JUSTIFICATION
1	$v \parallel w$	Given
2	$\angle 7 \cong \angle 11$	Given
3	$m\angle 7 = m\angle 11$	Defn. congruent
4	$m\angle 7 + m\angle 11 = 180$	Same-Side Int. Angles Thm.
5	$m\angle 11 + m\angle 11 = 180$	Substitution (3, 4)
6	$2(m\angle 11) = 180$	Simplified line 5
7	$m\angle 11 = 90$	Division Prop. Of Eq.
8	$m\angle 11 = m\angle 16$	Vertical Angles Thm.
9	$m\angle 16 = 90$	Substitution (7, 8)

3. Given: $l \parallel m$, $v \parallel w$
Prove: $\angle 1 \cong \angle 16$

	STATEMENT	JUSTIFICATION
1	$l \parallel m$	Given
2	$v \parallel w$	Given
3	$\angle 1 \cong \angle 3$	Corresp. Angles Post.
4	$\angle 3 \cong \angle 6$	Alt. Ext. Angles Thm.
5	$\angle 1 \cong \angle 6$	Transitive Prop. (3, 4)
6		
7		
8		

4. Given: $\angle 13 \cong \angle 4$, $l \parallel m$
Prove: $v \parallel w$

	STATEMENT	JUSTIFICATION
1	$\angle 13 \cong \angle 4$	Given
2	$l \parallel m$	Given
3	$\angle 13 \cong \angle 15$	Corresp. Angles Post.
4	$\angle 4 \cong \angle 15$	Transitive Prop. (1, 3)
5	$v \parallel w$	Converse of Alt. Ext. Angles Thm.
6		
7		
8		



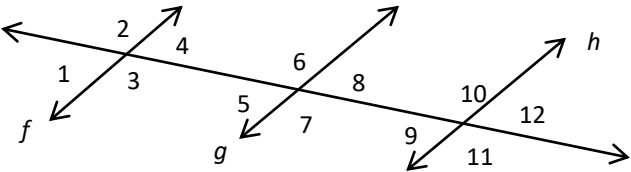
GEOMETRY PROOFS USING TRANSVERSALS
WARM-UP / EXIT TICKET A

Name: _____

Date: _____ Class: _____

Given: $\angle 11$ and $\angle 1$ are supplementary.

Prove: $\angle 3 \cong \angle 10$



	STATEMENT	JUSTIFICATION
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8		
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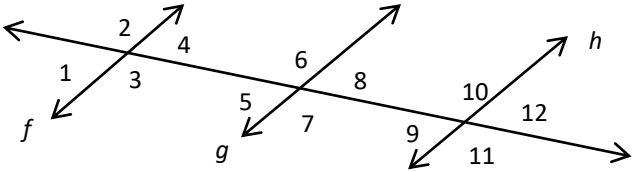
GEOMETRY PROOFS USING TRANSVERSALS
WARM-UP / EXIT TICKET B

Name: _____

Date: _____ Class: _____

Given: $\angle 12 \cong \angle 4$, $\angle 12 \cong \angle 5$

Prove: $m\angle 3 + m\angle 5 = 180^\circ$



	STATEMENT	JUSTIFICATION
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8		
9		
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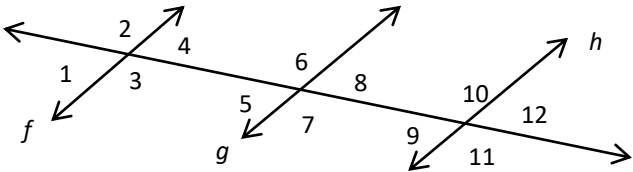
GEOMETRY PROOFS USING TRANSVERSALS
WARM-UP / EXIT TICKET A

Answer Key

Name: _____
Date: _____ Class: _____

Given:
Prove:

$\angle 11$ and $\angle 1$ are supplementary.
 $\angle 3 \cong \angle 10$



	STATEMENT	JUSTIFICATION
1	$\angle 11$ and $\angle 1$ are supplementary	Given
2	$f \parallel h$	Converse of Same-Side Exterior Angles Theorem
3	$\angle 3 \cong \angle 10$	Alternate Interior Angles Theorem
4		
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7		
8		
9		
10		

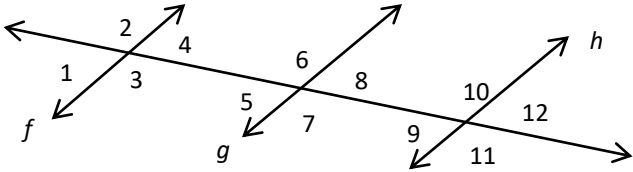
GEOMETRY PROOFS USING TRANSVERSALS
WARM-UP / EXIT TICKET B

Answer Key

Name: _____
Date: _____ Class: _____

Given:
Prove:

$\angle 12 \cong \angle 4, \angle 12 \cong \angle 5$
 $m\angle 3 + m\angle 5 = 180^\circ$



	STATEMENT	JUSTIFICATION
1	$\angle 12 \cong \angle 4$	Given
2	$\angle 12 \cong \angle 5$	Given
3	$\angle 4 \cong \angle 5$	Transitive Property (1, 2)
4	$f \parallel g$	Converse of Alternate Interior Angles Theorem
5	$m\angle 3 + m\angle 5 = 180$	Same-Side Interior Angles Theorem
6		
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8		
9		
10		

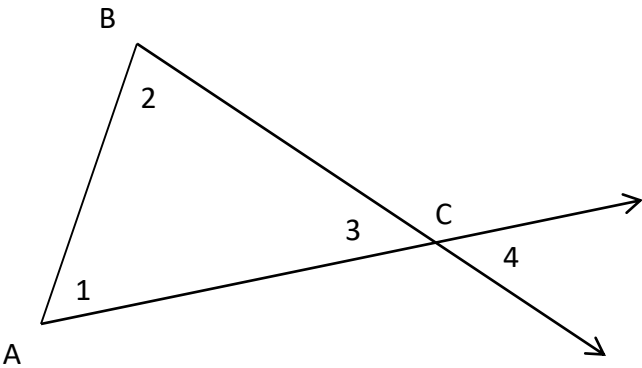


PROOFS WITH TRIANGLES

Name: _____
Date: _____ Class: _____

1. Given: $\angle 1 \cong \angle 4$
Prove: $\overline{AB} \cong \overline{BC}$

	STATEMENT	JUSTIFICATION
1		
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5		
6		

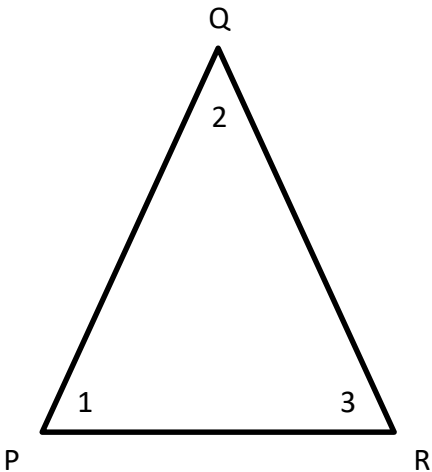


2. Given: $m\angle 1 + m\angle 2 = 140^\circ$
Prove: $m\angle 4 = 40^\circ$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		

3. Given: $AC = AB$
Prove: $m\angle 2 = m\angle 4$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		



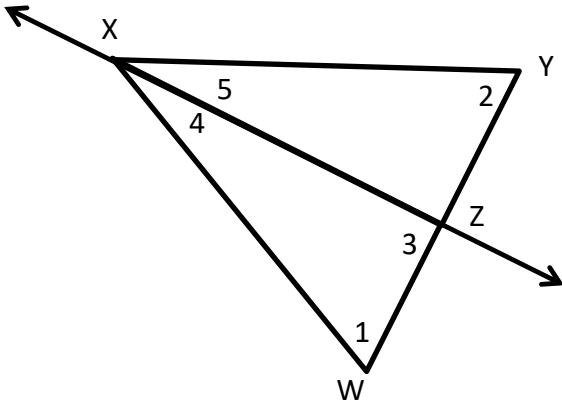
4. Given: $\angle 1 \cong \angle 3$, $\angle 1 \cong \angle 2$
Prove: $\overline{PQ} \cong \overline{PR}$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		



Name: _____

Date: _____ Class: _____



5. Given: $\angle 1 \cong \angle 2$, $XW = WY$

Prove: $WY = XY$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

6. Given: $WY = XW$

Prove: $m\angle 4 + m\angle 5 = m\angle 2$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
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7		
8		



PROOFS WITH TRIANGLES

Name: _____

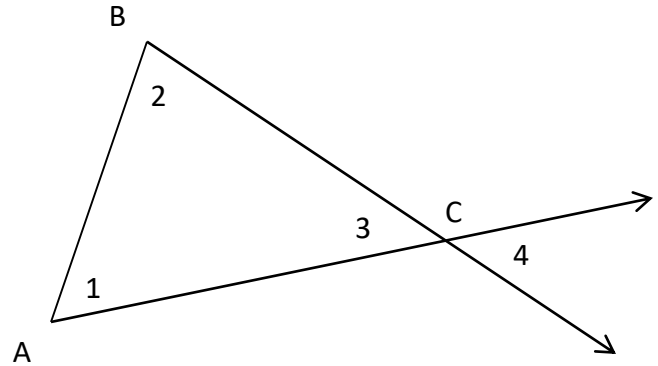
Date: _____ Class: _____

1. Given: $\angle 1 \cong \angle 4$

Prove: $\overline{AB} \cong \overline{BC}$

	STATEMENT	JUSTIFICATION
1	$\angle 1 \cong \angle 4$	Given
2	$\angle 3 \cong \angle 4$	Vertical Angles Theorem
3	$\angle 1 \cong \angle 3$	Transitive Prop. (1, 2)
4	$\overline{AB} \cong \overline{BC}$	Converse of Base Angles of Isos. Triangle Thm.
5		
6		

Answer Key



2. Given: $m\angle 1 + m\angle 2 = 140^\circ$

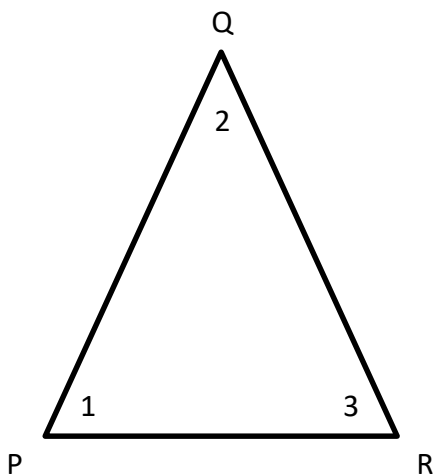
Prove: $m\angle 4 = 40^\circ$

	STATEMENT	JUSTIFICATION
1	$m\angle 1 + m\angle 2 = 140$	Given
2	$m\angle 1 + m\angle 2 + m\angle 3 = 180$	Triangle Sum Theorem
3	$140 + m\angle 3 = 180$	Substitution (1, 2)
4	$m\angle 3 = 40$	Subtraction Prop. of Eq.
5	$m\angle 3 = m\angle 4$	Vertical Angles Theorem
6	$m\angle 4 = 40$	Substitution (4, 5)

3. Given: $AC = AB$

Prove: $m\angle 2 = m\angle 4$

	STATEMENT	JUSTIFICATION
1	$AC = AB$	Given
2	$m\angle 2 = m\angle 3$	Base Angles of Isos. Triangle Thm.
3	$m\angle 3 = m\angle 4$	Vertical Angles Theorem
4	$m\angle 2 = m\angle 4$	Transitive Prop. (2, 3)
5		
6		



4. Given: $\angle 1 \cong \angle 3$, $\angle 1 \cong \angle 2$

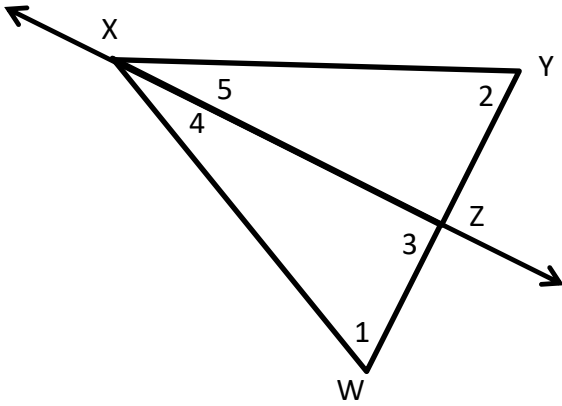
Prove: $\overline{PQ} \cong \overline{PR}$

	STATEMENT	JUSTIFICATION
1	$\angle 1 \cong \angle 3$	Given
2	$\angle 1 \cong \angle 2$	Given
3	$\angle 2 \cong \angle 3$	Transitive Prop. (1, 2)
4	$\overline{PQ} \cong \overline{PR}$	Converse of Base Angles of Isos. Triangle Thm.
5		
6		



Name: _____

Date: _____ Class: _____



5. Given: $\angle 1 \cong \angle 2$, $XW = WY$

Prove: $WY = XY$

	STATEMENT	JUSTIFICATION
1	$\angle 1 \cong \angle 2$	Given
2	$XW = WY$	Given
3	$XW = XY$	Converse of Base Angles of Isos. Triangle Thm.
4	$WY = XY$	Transitive Prop. (2, 3)
5		
6		
7		
8		

6. Given: $WY = XW$

Prove: $m\angle 4 + m\angle 5 = m\angle 2$

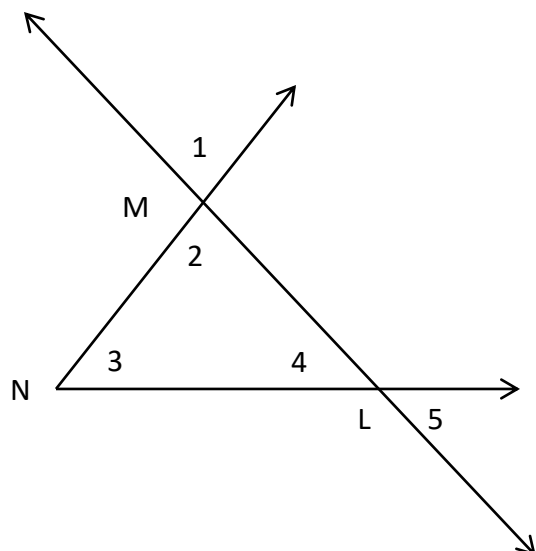
	STATEMENT	JUSTIFICATION
1	$WY = XW$	given
2	$m\angle WXY = m\angle 2$	Base Angles of Isos. Triangle Thm.
3	$m\angle 4 + m\angle 5 = m\angle WXY$	Angle Addition Post.
4	$m\angle 4 + m\angle 5 = m\angle 2$	Substitution (2, 3)
5		
6		
7		
8		



QUIZ: PROOFS USING SPECIAL ANGLE PAIRS AND TRIANGLES

Complete each proof.

Name: _____
Date: _____ Class: _____



1. Given: $\angle 1 \cong \angle 5$

Prove: $\overline{MN} \cong \overline{NL}$

	STATEMENT	JUSTIFICATION
1		
2		
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7		
8		

2. Given: $\overline{MN} \cong \overline{ML}$

Prove: $\angle 3 \cong \angle 5$

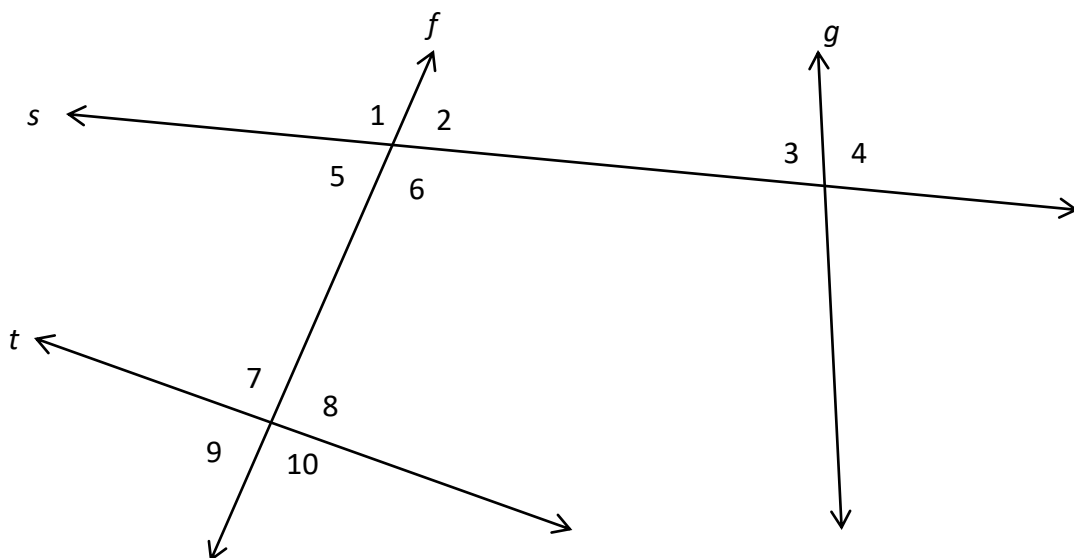
3. Given: $m\angle 2 = 90^\circ$, $m\angle 4 = 60^\circ$

Prove: $m\angle 3 = 30^\circ$

	STATEMENT	JUSTIFICATION
1		
2		
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4		
5		
6		
7		
8		

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		





4. Given: $f \parallel g$, $\angle 3 \cong \angle 7$

Prove: $s \parallel t$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
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6		
7		
8		
9		

5. Given: $\angle 8 \cong \angle 6$, $s \parallel t$

Prove: $m\angle 2 = 90^\circ$

	STATEMENT	JUSTIFICATION
1		
2		
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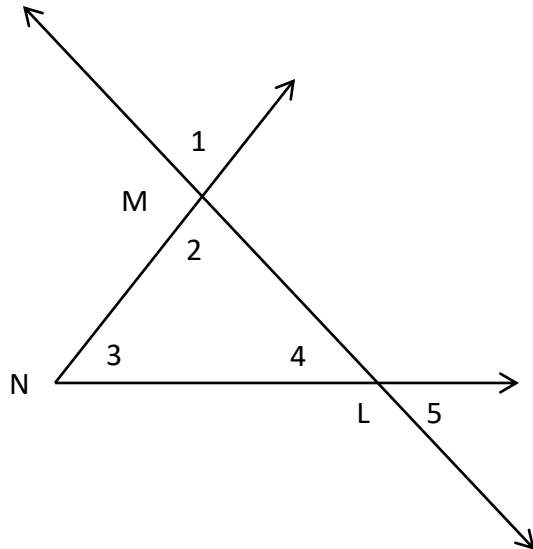


QUIZ: PROOFS USING SPECIAL ANGLE PAIRS AND TRIANGLES

Complete each proof.

Name: _____
Date: _____ Class: _____

Answer Key



1. Given: $\angle 1 \cong \angle 5$

Prove: $\overline{MN} \cong \overline{NL}$

	STATEMENT	JUSTIFICATION
1	$\angle 1 \cong \angle 5$	Given
2	$\angle 1 \cong \angle 2$	Vertical Angles Theorem
3	$\angle 4 \cong \angle 5$	Vertical Angles Theorem
4	$\angle 2 \cong \angle 5$	Transitive Prop. (1, 2)
5	$\angle 2 \cong \angle 4$	Transitive Prop. (3, 4)
6	$\overline{MN} \cong \overline{NL}$	Converse of Base Angles of Isos. Triangle Thm.
7		
8		

2. Given: $\overline{MN} \cong \overline{ML}$

Prove: $\angle 3 \cong \angle 5$

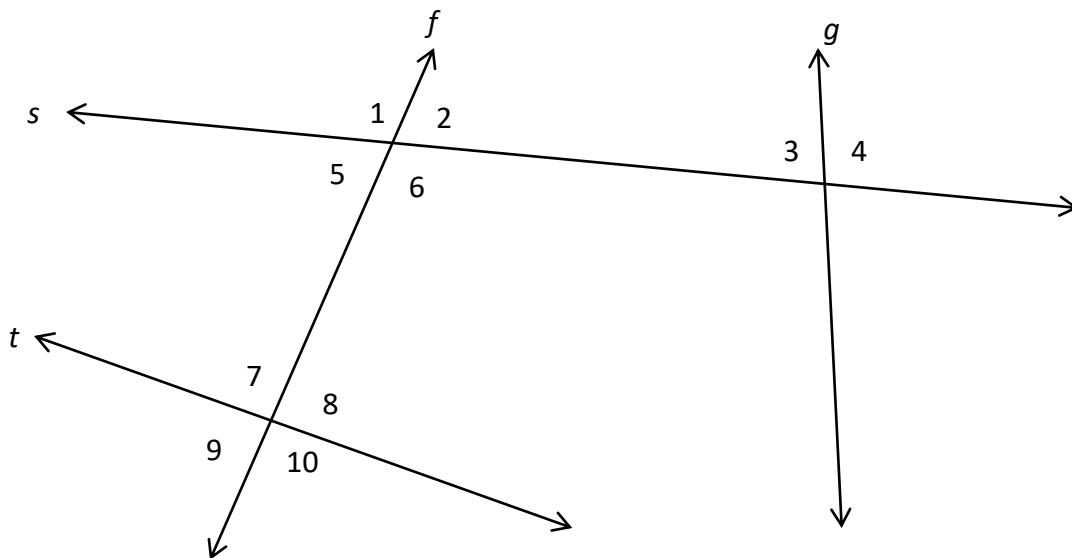
3. Given: $m\angle 2 = 90^\circ$, $m\angle 4 = 60^\circ$

Prove: $m\angle 3 = 30^\circ$

	STATEMENT	JUSTIFICATION
1	$\overline{MN} \cong \overline{ML}$	Given
2	$\angle 3 \cong \angle 4$	Base Angles of Isos. Triangle. Thm.
3	$\angle 4 \cong \angle 5$	Vertical Angles Theorem
4	$\angle 3 \cong \angle 5$	Transitive Prop. (2, 3)
5		
6		
7		
8		

	STATEMENT	JUSTIFICATION
1	$m\angle 2 = 90$	Given
2	$m\angle 4 = 60$	Given
3	$m\angle 2 + m\angle 4 + m\angle 3 = 180$	Triangle Sum Theorem
4	$90 + 60 + m\angle 3 = 180$	Substitution (1, 2, 3)
5	$150 + m\angle 3 = 180$	Simplified line 4
6	$m\angle 3 = 30$	Subtraction Prop. of Eq.
7		
8		





4. Given: $f \parallel g$, $\angle 3 \cong \angle 7$

Prove: $s \parallel t$

5. Given: $\angle 8 \cong \angle 6$, $s \parallel t$

Prove: $m\angle 2 = 90^\circ$

	STATEMENT	JUSTIFICATION
1	$f \parallel g$	Given
2	$\angle 3 \cong \angle 7$	Given
3	$\angle 3 \cong \angle 1$	Corresp. Angles Post.
4	$\angle 1 \cong \angle 7$	Transitive Prop. (2, 3)
5	$s \parallel t$	Converse of Corresp. Angles Post.
6		
7		
8		
9		

	STATEMENT	JUSTIFICATION
1	$\angle 8 \cong \angle 6$	Given
2	$s \parallel t$	Given
3	$m\angle 8 = m\angle 6$	Defn. congruent
4	$m\angle 8 + m\angle 6 = 180$	Same-Side Interior Angles Thm.
5	$m\angle 8 + m\angle 8 = 180$	Substitution (3, 4)
6	$2(m\angle 8) = 180$	Simplified line 5
7	$m\angle 8 = 90$	Division Prop. of Eq.
8	$m\angle 2 = m\angle 8$	Corresp. Angles Post.
9	$m\angle 2 = 90$	Substitution (7, 8)



TWO- COLUMN PROOF TEMPLATE

Name: _____

Date: _____ Class: _____

1. Given:

Prove:

2. Given:

Prove:

	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1			1		
2			2		
3			3		
4			4		
5			5		
6			6		
7			7		
8			8		

3. Given:

Prove:

4. Given:

Prove:

	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1			1		
2			2		
3			3		
4			4		
5			5		
6			6		
7			7		
8			8		



TWO-COLUMN PROOF TEMPLATE

Name: _____

Date: _____ Class: _____

Given:

Prove:

	STATEMENT	JUSTIFICATION
1		
2		
3		
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7		
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9		
10		

Given:

Prove:

	STATEMENT	JUSTIFICATION
1		
2		
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